Exploring new physics with proton interactions at the LHC

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Work done so far

Central exclusive production processes in proton-proton collisions.

Recent research has focused on lepton pair, WW/ZZ boson and top quark (tt) production, leading to:

- Precision Tests of the Standard Model
- Enhanced Sensitivity to New Physics: the ability to isolate exclusive events makes CEPs valuable for searching BSM;
- Forward Proton Tagging: the use of forward proton detectors - like PPS - improves event selection and provides more information about scattered protons, leading to more accurate measurements;

Why proton-proton collisions?

- Proton collisions at the LHC offer a unique environment for probing the Standard Model (SM)
- These collisions allow exploration of rare processes and potential new physics that are not accessible in lower-energy experiments.

Future work?

- Exploring the Anomalous Magnetic Moment of the Tau Lepton: unlike electrons and muons, leptons have higher mass and shorter lifetime, making it challenging to study the tau’s g-2 - requires high-energy experiments;
- Since taus are 3rd generation of leptons and heavier than electrons and muons, they may be even more sensitive to Beyond SM physics;

Thank you
for your attention

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